- Consider a string of m specified objects as a single unit
- Then we have $(n-m+1)$ objects. Permute this $(n-m+1)$ objects in $(n-m+1)!$ ways.
- Then permute the m specified objects between themselves in $m!$ ways.
- Finally, the answer is $m! \times (n-m+1)!$.

4.4.4 No two things are together (Gap method)

To obtain the number of permutations of n different objects when no two of k given objects occur together and there are no restrictions on the remaining $m = n-k$ objects, we follow the procedure as follows:

- First of all, arrange the m objects on which there is no restriction in a row. These m objects can be permuted in ${}^{m}P_{m} = m!$ ways.
- Then count the number of gaps between every two of m objects on which there is no restriction including the end positions. Number of such gaps will be one more than m that is $(m+1)$. In this $m+1$ gaps, we can permute the k objects in ${}^{m+1}P_{k}$ ways.
- Then the required number of ways are $m! \times {}^{(m+1)}P_{k}$.

Example 4.30 How many different strings can be formed together using the letters of the word "EQUATION" so that

(i) the vowels always come together?
(ii) the vowels never come together?

Solution:

(i) There are 8 letters in the word "EQUATION" which includes 5 vowels (E,U,A,I,O) and 3 consonants (Q,T,N). Considering 5 vowels as one letter, we have 4 letters which can be arranged in ${}^{4}P_{4} = 4!$ ways. But corresponding each of these arrangements, the vowels E,U,A,I,O can be put in ${}^{5}P_{5} = 5!$ ways.

Hence, by the rule of product required number of words is $4! \times 5! = 24 \times 120 = 2880$.

(ii) The total number of strings formed by using all the eight letters of the word "EQUATION" is

$$ {}^{8}P_{8} = 8! = 40320. $$

So, the total number of strings in which vowels are never together is the same as the difference between the total number of strings and the number of strings in which vowels are together is $40320 - 2880 = 37440$.

Example 4.31 There are 15 candidates writing an examination. 7 candidates are appearing for mathematics examination while the remaining 8 are appearing for different subjects. In how many ways can they be seated in a row so that no two candidates writing mathematics are together?

Solution:

Let us arrange the 8-non-mathematics candidates in ${}^{8}P_{8} = 8!$ ways. Each of these arrangements create 9 gaps. Therefore, the 7 candidates writing mathematics can be placed in these 9 gaps in ${}^{9}P_{7}$ ways.

$$ _O_1_O_2_O_3_O_4_O_5_O_6_O_7_O_8_ $$

By the rule of product, the required number of arrangements is

$$ 8! \times {}^{9}P_{7} = 8! \times \frac{9!}{2!} = \frac{8! \times 9!}{2!}. $$